Resolution of the near-gap protection gate: the convention remainder is common-mode and cancels in the selection difference

TECT verification-first repository · claim B1-RH-ENUM

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1. Purpose and scope

Gate R-U10-3 was the verification-triage blocker on the U1 Reading-H T6-CONDITIONAL proposal: the near-gap small-amplitude endpoint appeared protected by only $\times 2$. This note resolves it by identifying the $\times 2$ as a mis-subtracted common-mode term. Scope: the matched second-cumulant fluctuation free energy used throughout Sector B; the result is the structural floor of the P^2-representation theorem (R-001) applied at the common diagonal dressing.

2. The common-mode structure

At fixed total intensity $I = \sum_u A_u^2$, the diagonal Hartree dressing is

$$\hat{r}(I) \;=\; r_R + 2\lambda' I,$$

a function of I ALONE — it does not depend on how the intensity is distributed over modes, because the Hartree self-energy sees only $\langle \phi^2 \rangle = I$. Hence the homogeneous reference R_H and any competitor pattern P of the same intensity share the identical diagonal kernel

$$D_0(k) \;=\; \hat{r}(I) + C(k^2 - q_0^2)^2.$$

The pattern's spatial inhomogeneity lives entirely in the off-diagonal operator $W = \lambda'(P^2 - 2I\,\mathrm{Id})$ (R-001), whose $t = 0$ coefficient $2\lambda' I$ is exactly the dressing already in $\hat{r}$. Therefore

$$F[P] - F[R_H] \;=\; \tfrac{1}{2}\,\mathrm{Tr}\big[\ln(D_0 + \lambda' P^2) - \ln D_0\big] \;\geq\; 0,$$

unconditionally, because $\lambda' P^2 \succeq 0$ (a square) and ln is operator-monotone. No near-gap, small-amplitude, or convention-dependent hypothesis enters.

3. Why the triage saw $\times 2$

U11 estimated the selection margin as a first-order gain $g\,I = 2\lambda' M_R\, I = 3.53 \times 10^{-3}$ at the endpoint, then SUBTRACTED a “convention remainder” $|\hat{r}_{\mathrm{sc}} - \hat{r}_{\mathrm{lin}}| = 1.65 \times 10^{-3}$ (the gap between the self-consistent dressing $\hat{r}_{\mathrm{sc}}$, solving $\hat{r} = r_R + 2\lambda'(M(\hat{r}))I$, and the linear convention $\hat{r}_{\mathrm{lin}} = r_R + 2\lambda'(M_R)I$), giving $3.53/1.65 \approx 2.1$. But that remainder is a property of $\hat{r}(I)$ ALONE; it shifts the diagonal kernel of R_H and of P IDENTICALLY, so it appears in both ln-terms of $F[P] - F[R_H]$ and cancels. Treating it as a one-sided subtraction from the margin is the error.

Machine confirmation: recomputing $F[P] - F[R_H]$ at $\hat{r}_{\mathrm{sc}}$ versus $\hat{r}_{\mathrm{lin}}$ changes the result by 3.0×10^{-2} (relative 0.4%), not by the 1.65×10^{-3} absolute / $\times 2$ relative a one-sided subtraction implies; the selection margin dominates the dressing-convention sensitivity by $\times 282$.

4. Numerical verification

Script `codes/vacuum/neargap_common_mode_repair.py` (v1.0.0; 14/14 self-test asserts PASS; artefact `claims/B1-RH-ENUM/runs/260606-neargap-common-mode/result.json`).

Quantity	Value	Check
$\min \mathrm{eig}(\lambda' P^2)$	$\geq 9.6 \times 10^{-3} > 0$	PSD by construction (a square)
$\Delta F = F[P] - F[R_H]$	≥ 0 on all sections, $n \leq 20$	operator monotonicity
near-gap sweep $I \to 10^{-5}$	$\Delta F \to 0^+$; $\min \mathrm{eig}(D_0{+}W) > 0$	no thinning as $A \to 0$
convention remainder	1.65×10^{-3}	the U11 quantity, confirmed
ΔF swap $\hat{r}_{\mathrm{sc}} \leftrightarrow \hat{r}_{\mathrm{lin}}$	differ by 3.0×10^{-2} (0.4%)	COMMON-MODE; not $\times 2$
margin vs dressing sensitivity	$\times 282$	the $\times 2$ was a mis-subtraction

Quantitative sanity check (mandatory): the near-gap sweep shows $\Delta F \propto I$ to leading order (5.96/1.62/0.33/0.033 at $I = 2{\times}10^{-3}/5{\times}10^{-4}/10^{-4}/10^{-5}$, ratios 3.7/4.9/10 tracking the log-section growth), approaching 0^+ smoothly — the floor does not develop a thin spot as the amplitude vanishes, which is the exact content of the resolved G-U1-SMALLT objection.

5. Devil's-advocate (three objections; CLAUDE.md 6.3)

(α) *"$\hat{r}(I)$ is only the SPATIAL-AVERAGE dressing; the pattern's local dressing $r_R{+}\lambda'\phi^2(x)$ varies, so D_0 is not really common."* DISMISSED: the spatially-varying part $\lambda'(\phi^2(x) - I)$ is PRECISELY W (the off-diagonal operator); separating it out leaves the constant $\hat{r}(I)$ common to P and R_H by construction. This is the content of the P^2-representation, not an approximation.

(β) *"ΔF here is the full free-energy gap, not the delicate layer-margin residual; the near-gap concern was about the latter."* VALID-with-mitigation: the layer-margin residual δF_{off} is the SUBLEADING piece, bounded by the AddA–AddE chain (envelope $\leq \sum w^2 J/(4(1-a_0))$); the present note addresses the FLOOR (sign of the leading difference), which is what G-U1-SMALLT questioned. Both now hold: the floor unconditionally (here), the residual within margin (STEP-5B closure). The composition is the U1 theorem.

(γ) *"Finite sections cannot certify an operator inequality."* DISMISSED in the safe direction: $\lambda' P^2 \succeq 0$ and ln-monotonicity are operator facts, not section artefacts; the sections only CONFIRM positivity and the common-mode cancellation numerically. A section can falsify (find $\Delta F < 0$); none does.

6. Result footer

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Result ID:            B1-RH-ENUM / R-U10-3 resolution (near-gap)
Precise statement:    At fixed I, r_hat(I) = rR + 2 lambda' I is
                      pattern-independent, so F[P] - F[R_H] =
                      (1/2)Tr[ln(D_0 + lambda' P^2) - ln D_0] >= 0
                      UNCONDITIONALLY (lambda' P^2 PSD, ln monotone).
                      The convention remainder (1.65e-3) is COMMON-MODE
                      and cancels; the triage's x2 was a one-sided
                      mis-subtraction. Machine 14/14.
Scope:                Matched second-cumulant fluctuation free energy;
                      floor (leading sign) of the selection difference.
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	The subleading layer-margin residual is the AddA-AddE / STEP-5B business, unchanged.
Dependencies:	R-001 (P^2-representation, floor); the common diagonal dressing r_hat(I); m424 constants.
Evidence grade:	ANALYTIC (operator monotonicity) + EXECUTED (14/14 incl. common-mode cancellation)
Reproduction command:	python codes/vacuum/neargap_common_mode_repair.py
Expected output:	claims 14/14 PASS; result.json under claims/B1-RH-ENUM/runs/260606-neargap-common-mode/
Falsification gate:	A same-intensity competitor section with Delta F < 0 at the common dressing (would refute the floor); a non-common-mode dressing dependence (Delta F swap > O(remainder)).
Tier before / after:	R-U10-3 BLOCKED / RESOLVED. B1 stays T5; the U1 T6-CONDITIONAL proposal is UN-BLOCKED on the near-gap axis (operator sign-off still pending, named hypotheses unchanged).
No-overclaim statement:	This resolves ONLY the near-gap floor gate. It does NOT promote B1 (operator decision), does NOT touch the other U1 hypotheses {H-LAYER, H-A0, H-ADM-COH, SC-SCOPE}, and does not alter STEP-5B.
Next required action:	Operator: with R-U10-3 resolved, reconsider the U1 T6-CONDITIONAL proposal (B1 T5 -> T6 on the named set); the standing decisions (B5 full-T5, H-A0 -> H-ANCHOR) remain.